

The gene family of dehydration responsive element-binding transcription factors in grape (*Vitis vinifera*): genome-wide identification and analysis, expression profiles, and involvement in abiotic stress resistance

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Abstract The dehydration responsive element-binding (DREB) proteins play a critical role in plant development and abiotic stress-mediated gene expression. Therefore, they represent one of the most attractive regulons for breeding programs. However, no comprehensive summary of grapevine DREB family genes is available. During this study, 38 VvDREB members were identified from the entire grapevine genome and its expression sequence tag assembly. These were organized into the same subgroups, A1 through A6, as for *Arabidopsis* DREBs. The VvDREB genes were distributed in 15 out of 19 chromosomes in grapevine. Multiple sequence alignments were performed and a three-dimensional structure was created to demonstrate sequence conservation. Microarray analysis showed potential regulatory roles for VvDREBs in responses to various abiotic stresses, hormone treatments, berry ripening, exposure to light, and bud development. *Cis*-acting regulatory elements, such as W-box, MYB-binding site, and light-responsive elements, were the most frequently found in the putative promoter regions. Furthermore, microarray transcriptional profiling of grapevine plants that over-expressed

VvDREB23 revealed 248 up-regulated and 229 down-regulated genes, with fold-changes of >1.5 when compared with the empty vector control. Gene ontology classifications showed that different genes function in cellular glucan metabolism, lipid transport, the endomembrane system, cell wall structure, and other important metabolic and developmental processes, as well as in the regulation of molecular functions. Our report provides an overview and constitutes a foundation for further study of this VvDREB gene family. All the microarray data and transcription profiling of transgenic versus empty-vector control transformant grapevines were retrieved from the online resources.

Keywords DREB · Grapevine · Gene family · Gene expression · Gene ontology · *Vitis vinifera*

Abbreviations

AP2/ERF	APETALA 2/ethylene responsive element-binding factor
CBF	C-repeat binding factor
CRT	C-repeat
DRE	Dehydration responsive element
DREB	Dehydration responsive element-binding protein
VvDREB	<i>Vitis vinifera</i> dehydration responsive element-binding protein
TF	Transcription factor

Introduction

The dehydration responsive element-binding protein (DREB) is unique to the plant kingdom. As a member of the APETALA 2/ethylene responsive element-binding factor (AP2/ERF) superfamily, it is characterized by a

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conserved AP2 DNA-binding domain. It has been proved that they are major transcription factors (TFs) that respond to abiotic stresses by regulating downstream genes that contain a core DRE sequence (5'-A/GCCGAC-3') *cis*-acting element in their promoters [1–3]. The DREB subfamily can be further classified into six subgroups (A1–A6) based on their AP2 similarities [4]. In *Arabidopsis thaliana*, DREB1s are induced by cold [5, 6], while DREB2s are induced by drought, heat, and high salinity [4, 7, 8]. Other groups have not been studied as thoroughly. Differing from a DRE, the DREB3 orthologue *ZmABI4* binds to the CE-1-like sequence CACCG and regulates ABA- and sugar-responsive genes [9]. DREB4 is related to DREB1. Expression of the *Arabidopsis HARDY (HRD)* gene in rice improves water-use efficiency, i.e. the ratio of biomass produced to the amount of water used, by enhancing photosynthetic assimilation and reducing transpiration [10]. The *Arabidopsis DEAR1* in subgroup A5, accumulates in response to both pathogen infection and cold treatment, and suppresses the induction of DREB1/CBF family genes [11]. *Arabidopsis RAP2.4* in A6, plays a role in regulating light- and ethylene-mediated developmental processes and tolerance to drought. It is also involved in water homeostasis, regulating the functioning of multiple aquaporin genes [12]. All of these reports indicate that DREB TFs have complex and diverge roles in stress signaling pathways and other metabolic processes.

DREBs have been isolated and characterized from a wide range of species. Here are some recent advances, e.g. a proximal promoter region of *Arabidopsis DREB2C* confers tissue-specific expression under heat stress [13]. Oil palm DREB protein is responsive to treatment with ABA, ethylene, methyl jasmonate, drought, cold, and high salinity, and can also bind to an ABRE sequence [14]. Overexpression of a novel DREB TF from *Populus euphratica* improves drought and salt tolerances in *Arabidopsis* and *Lotus corniculatus* [15]. Drought-related DREB2-encoding genes exhibit nucleotide diversity in both wild and cultivated common bean [16]. Tomato *SlDREB* restricts leaf expansion and internode elongation by down-regulating key genes for gibberellin biosynthesis [17], whereas *OsAP21* increases salt/drought tolerance in transgenic *A. thaliana* [18]. Colinearity and similar expression patterns for rice DREB1s have revealed their functional conservation in the cold-responsive pathway [19].

Grapevine (*Vitis vinifera*) is one of the most widely cultivated, economically important woody plants. However, extreme temperatures, dehydration, and high salinity affect its growth and productivity. Previous research has shown that the *VvCBF4* TF enhances freezing tolerance in wine grapes [20]. Although Zhuang et al. [21] and Licausi et al. [22] have predicted 132 and 149 AP2/ERF

superfamily members, respectively, from the 8× assembled genome for *V. vinifera* ‘Pinot Noir’ PN40024 [23], no systematic and comprehensive information about the *VvDREB* subfamily is yet available. Moreover, a re-annotation of the *VvDREB* genes is required since the release of the 12× PN40024 genome that has been amended.

Here, we re-annotated the *VvDREB* members from that 12× grape genome as well as from the public expression sequence tag (EST) database. In addition, we investigated gene locations and structure, performed a phylogenetics analysis, identified conserved motifs, and created a three-dimensional model. Using publicly available microarrays, we examined expression profiles under various abiotic stresses, as well as during fruit ripening and bud development, and in response to hormone and light treatments. Finally, we used a microarray of grapevine that over-expresses *VvCBF4* (termed *VvDREB23* here) to analyze differently expressed genes towards control and make gene ontology (GO) classifications. We believe that our results can be used to understand the regulative mechanisms of *VvDREB1s* in biological processes, cellular components, and molecular functions.

Materials and methods

Identification of *VvDREBs* and sequence refinement

The *V. vinifera* 12× peptidic FASTA file was downloaded from Ensembl Plants website, (http://plants.ensembl.org/Vitis_vinifera/Info/Index?db=core). The following steps were then taken:

Step 1 Use all 29,927 *V. vinifera* peptide sequences as queries to conduct a BLASTP algorithm search against a local *A. thaliana* protein database (TAIR10_pep_20101214, 35,386 aa), setting the *E* value cutoff as $1e-3$. If the best hit was one of the 56 *AtDREBs* [4], then that query sequence was considered a candidate DREB member. In order to include sequences that may not have been predicted/correctly predicted in the genome FASTA file, we moved to the next step.

Step 2 Use candidate sequences from Step 1 as queries to perform a TBLASTN search of all *V. vinifera* unigenes (<http://www.ncbi.nlm.nih.gov/unigene>; 22,101 records), setting the *E* value as $1e-6$. All of the unigene sequences were collected and duplications were deleted.

Step 3 Use the sequences from Step 2 to perform a BLASTX of the same *A. thaliana* database in Step One and screen out the unigenes that encode *VvDREBs*. Afterward, gather and align the sequences from Steps 1 and 3, and reject the redundant records.

The coding sequences of *VvDREBs* identified from the genome data were then validated and revised by dbUnigene and dbEST, using a MegaBLAST tool with criteria that

included identity >95 %, aligned length >100 bp, and mismatches of <20. Finally, these proteins were submitted to the NCBI batch web CD search tool (<http://www.ncbi.nlm.nih.gov/Structure/bwrpsb/bwrpsb.cgi>), Pfam (<http://pfam.sanger.ac.uk/>), and SMART (<http://smart.embl-heidelberg.de/>) to confirm the presence and completeness of the AP2 domain.

Sequence alignment, motif identification, phylogenetic tree, and three-dimensional structure construction

Multiple sequences were aligned with CLC Workbench version 6 (www.clcbio.com). The Multiple Expectation maximization for Motif Elicitation tool (MEME version 4.9.0; <http://meme.nbcr.net/meme/cgi-bin/meme.cgi>) was used to search the motifs; minimum and maximum widths, and the maximum number of motifs were set at 6, 20, and 12, respectively [24]. We reconstructed the phylogenetic tree by the neighbor-joining method in MEGA 5.0 software, using the Jones–Taylor–Thornton (JTT) model, and setting the number of bootstrap replications at 1,000 [25]. Sequence conservation of the 3-D *VvDREB* AP2 domain was performed in Chimera 1.7s [26]. The nuclear magnetic resonance structure of AP2 domains was obtained from PDB ID; 1GCC, 2GCC, 3GCC, and the AP2 domain of *VvDREB34* were based on the PDB models [27]. Afterward, the structure was color-coded to indicate the conservation of amino acids among all *VvDREBs*. GFF3 file was also downloaded from EnsemblPlants website to determine chromosomal location and gene structure of *VvDREBs* [28].

Analyses of *VvDREB* expression and promoter activity

Robust Multi-array Average normalized microarray expression data was downloaded from Plant Expression Database. The analysis covered 14 microarray experiments, including those involving various abiotic stresses, hormone applications, berry ripening, light responses, and bud development. The Affymetrix GeneChip 16K *V. vinifera* (Grape) Genome Array was used to evaluate experiments VV1, 2, 3, 10, 11, 15, 16, 17, 29, and 40, while the *V. vinifera* GrapeGen (Custom Affymetrix Grape) Array was used with experiments VV33, 36, 42, and 43. A BLAST search was conducted to find corresponding probes for a given *VvDREB* gene. Probe sequences were obtained from the Affymetrix website (<http://www.affymetrix.com/>), and Multiexperiment Viewer (MeV) software [29] was used to produce a heatmap. Finally, 1.5 kb of the upstream sequences of the ATG initial codon for each *VvDREB* gene was submitted to PlantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) [30].

Genome-wide changes in gene expression and gene ontology analysis of the *VvDREB* overexpressor

Expression data from 35S::*VvCBF4* (*VvDREB23*)-over-expressing grapevines (VV31) was downloaded and imported into MeV. The Limma package was used to find genes differentially expressed between the transgenic and control plants [31]. Genes fulfilling the criteria of a p value ≤ 0.05 and a fold-change ≥ 1.5 when means from the two groups were compared were considered to have significantly altered expression. The GO classification of genes that had significantly changed expression was performed with the agriGO tool (<http://bioinfo.cau.edu.cn/agriGO/>) [32]. The singular enrichment analysis (SEA) method was used to identify enriched GO terms. The grape Affymetrix Genome Array was selected as background, and the Fisher method for statistical tests was used with a significance level set at $p \leq 0.05$.

Results

Identification and nomenclature of *VvDREBs*

We identified 38 *VvDREBs* and 0 new from the $12\times V. vinifera$ genome (29,927 genes) and the grapevine dbUni-gene (22,101 genes), respectively. These 38 *VvDREBs* were designated *VvDREB01* to *VvDREB38* according to their location, top to bottom, on Chromosomes 1 through 19 (Table 1). More information and comparison with former studies (Zhuang et al. [21]; Licausi et al. [22]) are listed in Supplementary Table S1.

Sequence conservation, additional motifs, and phylogenetic analysis

Multiple sequence alignment showed that the 38 *VvDREBs* share a conserved DNA-binding AP2 domain (Fig. 1a). Our 3-D structure of all *VvDREB* AP2 domains comprised a three-stranded anti-parallel beta sheet and an alpha helix packed approximately parallel to the beta sheet (Fig. 1b, c). The color bar indicated the gradient of conservation from low to high. It was notable that the 6th arginine (R6), the 14th valine (V14), the 16th aminoglutaric acid (E16), and the 27th tryptophan (W27) are 100 % identical (Fig. 1b). Allen et al. [27] had reported earlier that arginine and tryptophan residues in the beta sheet have contact with eight of the nine consecutive base pairs in the major groove, while also binding to the sugar phosphate backbones. Moreover, Sakuma et al. [4] have shown that V14 has important roles in DNA-binding specificity.

The MEME motif demonstrated how the *VvDREB* proteins are distributed (Fig. 2). Of the 12 motifs found here,

Table 1 Grapevine DREB genes

Name	Gene ID	Length	Best hit	Introns	Location	From	To	Subgroup
VvDREB01	VIT_02s0025g01360	264	AT1G64380	2	Chr2	1,274,105	1,275,209	A-6
VvDREB02	VIT_02s0025g04440	186	AT1G12630	0	Chr2	3,903,010	3,903,835	A-4
VvDREB03	VIT_04s0008g02230	191	AT1G19210	0	Chr4	1,834,035	1,834,800	A-5
VvDREB04	VIT_04s0008g03400	241	AT5G11590	0	Chr4	2,818,241	2,819,492	A-4
VvDREB05	VIT_05s0029g00140	404	AT4G13620	1	Chr5	14,296,111	14,297,784	A-6
VvDREB06	VIT_06s0061g01390	253	AT5G51990	0	Chr6	19,216,576	19,217,622	A-1
VvDREB07	VIT_06s0061g01400	211	AT5G51990	1	Chr6	19,254,403	19,255,122	A-1
VvDREB08	VIT_07s0031g00190	152	AT2G23340	0	Chr7	16,404,262	16,405,010	A-5
VvDREB09	VIT_08s0007g03790	212	AT5G51990	0	Chr8	17,765,904	17,766,542	A-1
VvDREB10	VIT_08s0007g03810	199	AT2G36450	0	Chr8	17,783,389	17,784,006	A-4
VvDREB11	VIT_08s0007g08150	287	AT3G57600	0	Chr8	21,518,702	21,520,332	A-2
VvDREB12	VIT_09s0002g03940	235	AT5G11590	0	Chr9	3,665,089	3,665,967	A-4
VvDREB13	VIT_11s0016g00660	205	AT1G19210	0	Chr11	646,736	647,692	A-5
VvDREB14	VIT_11s0016g00670	150	AT5G21960	0	Chr11	654,774	655,295	A-5
VvDREB15	VIT_11s0016g02140	147	AT4G25480	3	Chr11	1,758,493	1,761,259	A-1
VvDREB16	VIT_11s0016g03350	266	AT5G11590	0	Chr11	2,729,375	2,730,729	A-4
VvDREB17	VIT_12s0059g00280	465	AT2G20880	0	Chr12	5,139,472	5,141,309	A-6
VvDREB18	VIT_13s0067g01400	306	AT2G40220	0	Chr13	792,365	793,285	A-3
VvDREB19	VIT_13s0067g01960	355	AT2G40340	1	Chr13	1,081,303	1,084,157	A-2
VvDREB20	VIT_14s0006g02290	236	AT3G60490	0	Chr14	19,671,450	19,672,638	A-4
VvDREB21	VIT_15s0021g02110	188	AT3G60490	0	Chr15	12,934,820	12,935,386	A-4
VvDREB22	VIT_15s0046g00310	207	AT1G01250	0	Chr15	17,321,737	17,322,484	A-4
VvDREB23	VIT_16s0100g00380	218	AT5G51990	0	Chr16	15,813,825	15,814,917	A-1
VvDREB24	VIT_16s0100g00390	164	AT5G52020	0	Chr16	15,832,474	15,832,968	A-4
VvDREB25	VIT_16s0100g00400	237	AT5G52020	0	Chr16	15,837,953	15,838,931	A-4
VvDREB26	VIT_18s0001g03240	233	AT1G21910	1	Chr18	3,219,544	3,220,699	A-5
VvDREB27	VIT_18s0001g05250	367	AT1G78080	0	Chr18	4,220,268	4,222,313	A-6
VvDREB28	VIT_18s0001g10150	141	AT5G67190	0	Chr18	8,500,950	8,501,729	A-5
VvDREB29	VIT_18s0001g13320	201	AT1G75490	0	Chr18	11,347,342	11,348,103	A-2
VvDREB30	VIT_18s0089g01030	174	AT1G77200	0	Chr18	28,924,140	28,924,766	A-4
VvDREB31	VIT_18s0001g05850	177	AT1G33760	0	Chr18_random	4,529,837	4,530,370	A-5
VvDREB32	VIT_18s0001g05890	182	AT1G71450	0	Chr18_random	4,541,824	4,542,582	A-4
VvDREB33	VIT_19s0014g03180	403	AT2G20880	0	Chr19	3,327,403	3,328,786	A-6
VvDREB34	VIT_00s0341g00070	239	AT2G44940	0	ChrUn	24,451,343	24,452,062	A-4
VvDREB35	VIT_00s0341g00080	225	AT3G60490	0	ChrUn	24,460,830	24,461,680	A-4
VvDREB36	VIT_00s0632g00010	230	AT2G44940	0	ChrUn	33,261,288	33,261,979	A-4
VvDREB37	VIT_00s0662g00030	229	AT1G78080	1	ChrUn	33,763,132	33,765,774	A-6
VvDREB38	VIT_00s0662g00040	374	AT1G78080	0	ChrUn	33,765,775	33,767,230	A-6

numbers 1–4 are components of the AP2 domain, occurring 37 or 38 times among the 38 *VvDREBs*. In addition, we identified group-specific motifs, e.g. motif 7 ([NS]H[IL][TL]P[AS][NQ][IV][LH]QIQAAQ[IQ]Q[FLQ]QQ[QH]), motif 8 ([NT][MQ]YPD[FG]CSTSTTH[VM][FI]SQGF), motif 9 ([QW]Q[QR]Q[HQ][ANT][LP]NFL[GS]PK[AP]IPMKQV), which are simultaneously present in three members of A6 (Fig. 2; Table S2). Motif 6 ([VD]D[E-D][ED][AT][FL][FL][DN][ML]P[DNG]L[LF]V[DNS][M-V]AEG[LMF]) occurs in most *VvDREB1s* and *VvDREB4*.

By contrast, Motif 12, which shows highly conserved regular expression of '[HPQ]K[RK][KR][AT]GRKKF[RK]', occurs in only four A1 members (Fig. 2; Table S2).

Chromosomal location and gene structure

The *VvDREBs* are distributed unevenly across 15 chromosomes of the grapevine genome (Table 1). Chromosome 18 has the most—five genes—while

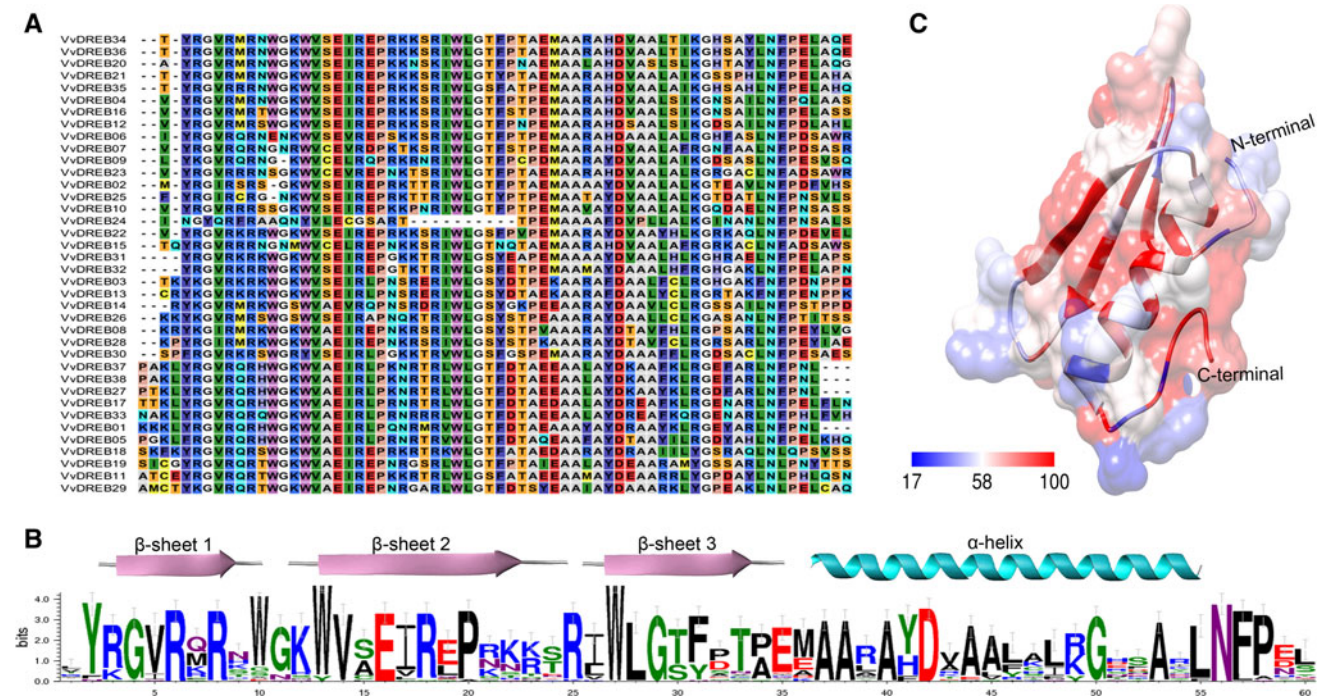


Fig. 1 **a** Alignment of AP2 domain of 35 *VvDREB* proteins, performed via CLC Combined Workbench version 6 (www.clcbio.com). **b** Sequence logo of AP2 domain, generated by WebLogo (<http://weblogo.threeplusone.com>); β -sheet and α -helix appear at top

of corresponding amino acid sequences. **c** Three-dimensional structure of AP2 DNA-binding domain; color key indicates percentage of conservation levels, from low (blue) to high (red)

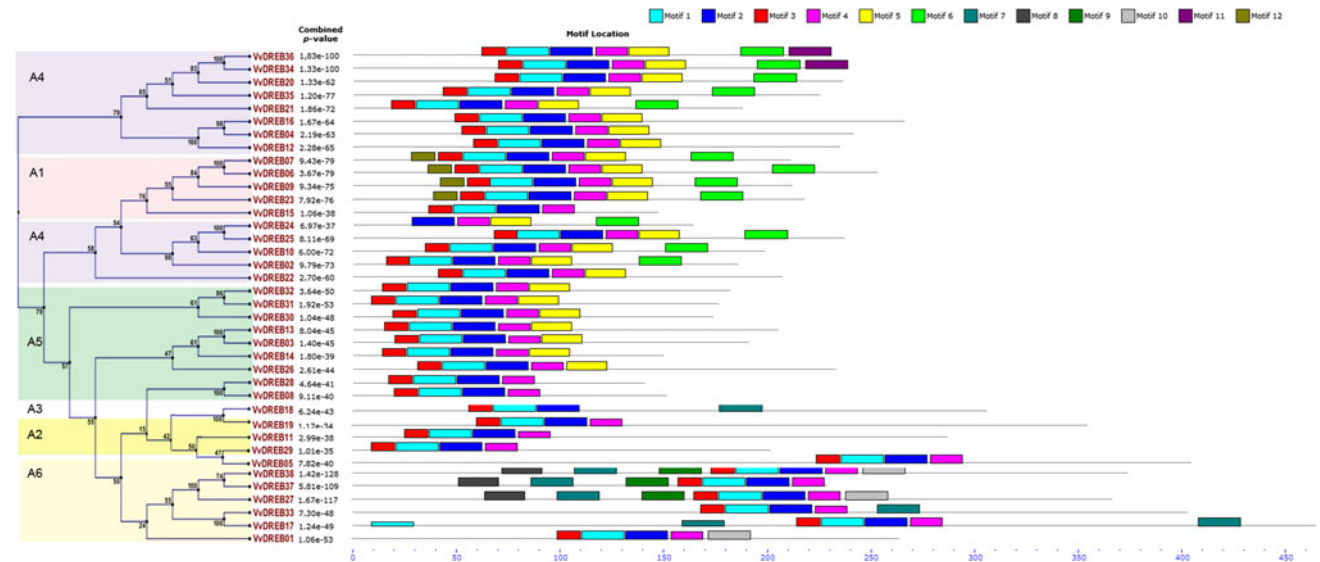


Fig. 2 **a** Phylogenetic tree constructed by MEGA5 software and motif locations identified by MEME program. Height of motif “block” is proportional to $-\log(p \text{ value})$; combined p values are shown in middle. **b** Sequence logos and detailed information for each motif

chromosomes 11 contain four genes. Chromosomes 1, 3, and 10 carry no DREB genes. Both *VvDREB31* and *VvDREB32* are located on random Chromosome 18, while *VvDREB34* through 38 are found on an undetermined chromosome.

Most *VvDREBs* have no introns. As a result, five genes (*VvDREB05*, 07, 19, 26, and 37) have one intron each, and

VvDREB15, and *VvDREB01* have 3 and 2 introns, respectively.

Expression profiles of *VvDREBs*

We obtained microarray data from two Affymetrix grapevine platforms—the Affymetrix GeneChip 16K *V. vinifera*

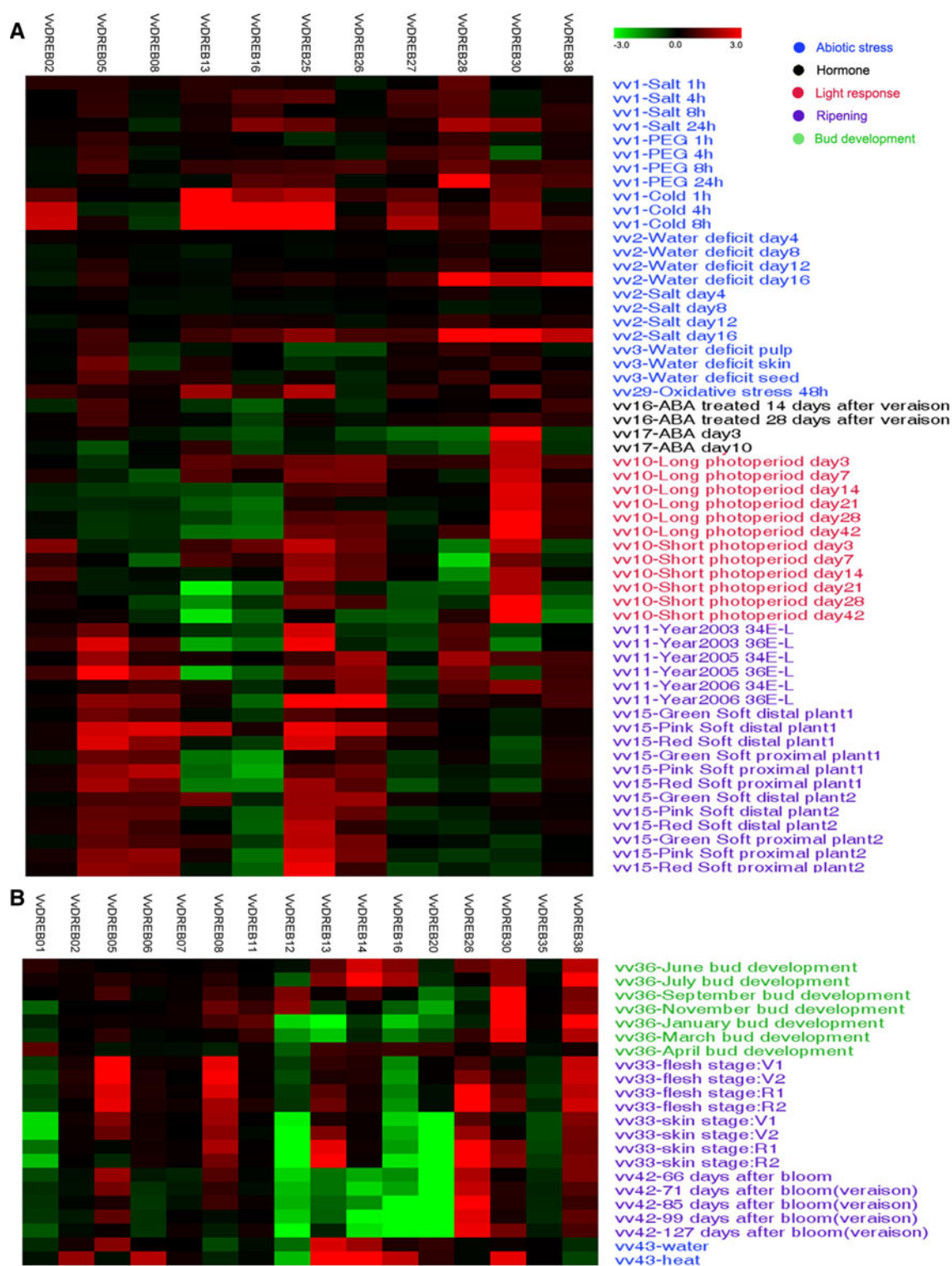


Fig. 3 Expression profiles of *VvDREBs* identified by using two Affymetrix platforms: **a** GeneChip 16K Grape Genome Array, and **b** Custom Affymetrix Grape Array. Intensity of expression is defined in colored bar at top of chart. Genes are presented in numerical order,

expressed under conditions listed on right; colored fonts indicate treatment types, as defined with colored dots at top of chart. The scale bar represents the log 2 value from -3.0 to 3.0

(Grape) Genome Array (Fig. 3a) and the *V. vinifera* GrapeGen (Custom Affymetrix Grape) Array (Fig. 3b)—which enabled us to identify 11 and 17 *VvDREB* genes, respectively. In all, 19 *VvDREB* expression profiles can be investigated under various experimental treatments, including abiotic stress [33–36], ABA [37], light response [38], fruit ripening [39–42], and the process of bud development [43] (Table S3).

The experiments with drought, salt, cold, heat, and oxidative stresses included short-term assays conducted under high salinity, polyethylene glycol (PEG), and cold (5 °C) that lasted for 0, 1, 4, 8, or 24 h (Table S3). Plants of ‘Cabernet Sauvignon’ grape were grown in a hydroponic drip system. From the salt treatment, seven genes showed obviously increased expression (>1.5-fold change) (Fig. 3a; Table S3). These included *VvDREB05*, 27, and 38 in Subgroup A6; *VvDREB16*, 25, and 30 in A4; and *VvDREB28* in A5 (Fig. 3a; Table S3). Among these, *VvDREB16*, 25, and 28 had relatively higher expression that peaked at hour 24 after treatment, with respective fold-changes of 2.8, 2.3, and 3.9 (Fig. 3a; Table S3). We found it interesting that *VvDREB30* was slightly down-regulated in the first 8 h of salt treatment before its expression increased threefold by hour 24 (Fig. 3a; Table S3). This gene showed a similar pattern in response to PEG treatment, first being down-regulated before being up-regulated (Fig. 3a; Table S3). Another eight *VvDREBs* also had fold increases >1.5, including *VvDREB28*, for which expression was enhanced by about 8.0-fold over the control after 24 h (Fig. 3a; Table S3). After treatment at 5 °C, 6 of the 11 *VvDREBs* showed over threefold increases; most remarkably, *VvDREB13* in A5, and *VvDREB16* and 25 in A4 were up-regulated over the control by 22-, 14-, and 16-fold, respectively (Fig. 3a; Table S3).

Long-term salt and water stress experiments [33] were conducted in the greenhouse, using potted ‘Cabernet Sauvignon’ vines. Compared with results from the short-term trials, the expression of most genes remained almost unchanged for the first 12 days. However, at day 16 after irrigation was withheld, *VvDREB28* in A5, *VvDREB30* in A4, and *VvDREB38* in A6 showed significant upregulation (14-, 5-, and 7-fold, respectively) (Fig. 3a; Table S3). At that point in the experiment, it also appeared that expression by *VvDREB05*, 16, 25, and 27 was just beginning to increase (Fig. 3a; Table S3). During long-term saline treatments, *VvDREBs* also showed little change in the first 12 days before being obviously up-regulated at day 16 (Fig. 3a; Table S3). These included *VvDREB28*, 30, and 38, which were up-regulated about 13-, 9-, and 5-fold, respectively, over the control (Fig. 3a; Table S3). Microarray analysis that compared among samples of pulp, skin, and seeds from berries of ‘Cabernet Sauvignon’ plants under water deficit showed that *VvDREB05* and 30 were

induced and had the greatest expression in the skin (Fig. 3a; Table S3). However, *VvDREB08*, 25, and 26 were down-regulated in both pulp and skin (Fig. 3a; Table S3).

Within that same study, the effects of oxidative stress were examined when micropropagated *V. vinifera* were transferred to ex vitro high-light conditions for 48 h [35]. There, six *VvDREB* genes showed an increase in expression of >1.5-fold, with *VvDREB13*, 25, and 30 having fold-changes of 3.6, 4.3, and 2.9, respectively. Under heat stress (Fig. 3b) [36], *VvDREB06* in A1; *VvDREB02*, 16, and 30 in A4; and *VvDREB13* and 14 in A5 were up-regulated by >3-fold. In particular, *VvDREB13* and 14 showed increases in expression that were >123-fold and 37-fold, respectively, when compared with the control (Fig. 3b; Table S3).

Two investigations have been reported concerning the expression profiles of *VvDREBs* under ABA treatment [37]. In the first, VV16, field-grown grapevine clusters of ‘Cabernet Sauvignon’ were sprayed with a 400 mg L⁻¹ ABA solution at véraison (time of blooming) whereas, in the second experiment, VV17, berries from plants in the same field were excised and cultured in vitro with 700 mg L⁻¹ ABA. When the fruit skins were sampled from both VV16 and VV17, expression tended to follow approximately the same patterns with *VvDREB14*, 16, 21, and 23 being down-regulated by as much as twofold (Fig. 3a; Table S3). We found it noteworthy, however, that *VvDREB30* in A4, was up-regulated by 16.7-fold on day 3 and by 4.5-fold on day 10. Although *VvDREB38* was up-regulated in experiment VV16 but down-regulated in VV17, this may have been due to the use of different ABA concentrations (Fig. 3a; Table S3).

VvDREBs were also responsive to different photoperiods [38] (Fig. 3a; Table S3). *VvDREB38* was up-regulated under long days but down-regulated under short days (Fig. 3a; Table S3). Some dramatic responses to a short photoperiod were also found, e.g. expression was down-regulated by 7.0- and 6.5-fold for *VvDREB13* on days 21 and 42, respectively, and by 5.6-fold for *VvDREB28* on day 7 (Fig. 3a; Table S3). By contrast, *VvDREB30* showed higher expression when compared with the control (2.4- to 26.0-fold), whether the photoperiod was long (15 h) or short (13 h) (Fig. 3a; Table S3).

Microarray data from four experiments (VV11, 15, 33, and 42) of berry development and ripening were used to investigate whether *VvDREBs* are involved in this important biological process. As shown in the heatmap (Fig. 3), results were consistent between VV11 and 15, where *VvDREB02* and *VvDREB38* had little response during the ripening period while expression of *VvDREB05*, 08, 16, and 25 was increased during that time (Fig. 3a; Table S3). *VvDREB16*, 13, and 26 underwent significant downregulation at specific time points (Fig. 3a; Table S3). Furthermore, positioning on the vine affected expression, with

genes in proximal berries showing greater downregulation than those in distal berries (Fig. 3a; Table S3). Experiment VV33 demonstrated that *VvDREB02*, *06*, *07*, *11*, and *14* were non-responsive in both flesh and skin during fruit developmental stages (Fig. 3b; Table S3). By comparison, *VvDREB05*, *08*, and *38* were up-regulated during that process, with expression being higher in the flesh than in the skin (Fig. 3b; Table S3). We also found it interesting that *VvDREB13*, *26*, and *30* were more likely to be up-regulated in the ripening stages (R1 and R2) than in the stages of 50 % véraison (V1) and 100 % véraison (V2) (Fig. 3b; Table S3). It is important to note that *VvDREB01*, *VvDREB12*, and *VvDREB16* were markedly down-regulated, especially in the skin, throughout these developmental stages (Fig. 3b; Table S3). In particular, *VvDREB20* was strongly down-regulated (8- to 14-fold) only in the berry skin while its expression in the flesh remained unchanged (Fig. 3b; Table S3). A similar trend was observed in experiment VV42, where this gene was down-regulated by 90- to 125-fold from day 85 to 127 after véraison (Fig. 3b; Table S3). Expression of *VvDREB12*, *14*, and *16* was also significantly repressed by 8- to 30-fold during this period (Fig. 3b; Table S3).

While the grapevine buds were developing, the expression of *VvDREB02*, *06*, *07*, and *35* remained almost unchanged (Fig. 3b; Table S3). *VvDREB13*, *14*, *16*, *30*, and *38* were all up-regulated in June and July, but their patterns of expression became diverse afterward (Fig. 3b; Table S3). For example, *VvDREB13*, *14*, and *16* were down-regulated from September to March (especially January through March) while transcripts of *VvDREB30* and *VvDREB38* continued to rise during that period. In fact, *VvDREB30* expression was sharply increased, by 42- and 33-fold, in September and November, respectively (Fig. 3b; Table S3). Of all the *VvDREBs* examined in that experiment [43], the expression pattern for *VvDREB12* was unique, first being down-regulated twofold in July and then up-regulated about threefold from September to November, before being down-regulated again from January to April. In particular, its expression in January decreased by approximately sixfold (Fig. 3b; Table S3).

Cis-acting regulatory elements in the promoter region of *VvDREBs*

We submitted 1.5-kb-long genomic DNA sequences upstream of the ATG start codon to the PlantCARE database in order to search for *cis*-acting elements (Fig. 4). *Arabidopsis thaliana* promoter sequences were used as our reference. Among the motifs identified here, MBS (MYB binding site involved in drought-inducibility), a W-box, ABRE (involved in the abscisic acid response), and several light-responsive elements (AE-box, ATCT-motif, GA-

motif, CAG-motif, Gap-box, GAG-motif, GT1-motif, G-box, GATA-motif, and TCT-motif) are the most frequently found types of *cis*-elements. Other regulatory elements, such as JERE (jasmonate ethylene-responsive element), DRE (involved in responses to dehydration, low temperatures, salt), CAT-box (related to meristem expression), and HD-Zip (involved in the control of leaf morphology) are also present.

Analysis of differential expression and GO enrichment in *VvDREB19*-overexpressing grapevine

Microarray-based mRNA expression profiling of 35S::*VvDREB23* (*VvCBF4*) in a line of transgenic grapevines was conducted, and an empty-vector transformant was used as the control [20]. Initial results from the LIMMA algorithm indicated that 992 genes are differentially expressed (adjusted *p* value ≤ 0.05), which accounts for 6 % of all genes (Fig. 5a). Subsequent analysis of the fold-changes reduced this number to 477 genes, which fit our expression criteria (fold change ≥ 1.5). Of those, 248 are up-regulated and 229 are down-regulated (Table S4).

Corresponding probes were submitted to AgriGO separately for GO-term classification and enrichment (Fig. 5b). Of the 248 up-regulated genes, 196 have been annotated and 22 significant GO terms are now enriched. ‘Lipid transport’ is the most significant GO term in the annotation of biological processes followed by processes for ‘cellular glucan metabolism’, ‘carbohydrate metabolism’, and ‘polysaccharide metabolism’. With regard to molecular function ontology, the most significant term is ‘structural constituent of cell wall’ followed by ‘hydrolase activity, acting on glycosyl bonds’, ‘enzyme inhibitor activity’, ‘carboxylesterase activity’, ‘enzyme regulator activity’, and ‘copper ion binding’. Within the classification of cellular components, ‘cytoplasmic membrane-bounded vesicle’ is most frequent followed by ‘cell wall’, ‘endomembrane system’, and ‘apoplast’ (Fig. 5b).

The 196 genes with down-regulated transcripts have been annotated with 23 significant GO terms (Fig. 5b). The most common category in biological processes is ‘phenylpropanoid biosynthesis’ followed by ‘response to stress’, ‘phenylpropanoid metabolism’, ‘aromatic compound biosynthesis’, ‘cellular amino acid derivative biosynthesis’, ‘response to stimulus’, ‘alkene biosynthesis’, ‘lignin biosynthesis’, and ‘response to heat’. In the molecular function category, the order is ‘monooxygenase activity’, ‘heme-binding’, ‘tetrapyrrole-binding’, ‘oxidoreductase activity’, ‘iron ion-binding’, ‘heat shock protein-binding’, ‘vitamin-binding’, and ‘transferase activity, transferring hexosyl groups’. No significant GO terms were found in the cellular component annotation.

All statistically significant terms are shown in a GO hierarchical graph (Fig. 6). This presents the changes in



biological processes, molecular functions, and cellular components from a systemic perspective.

In this study, we have identified 38 *VvDREBs* in the 12 × v1 assembled grapevine genome. The DREB gene family

Compared with other plants, grapevine has significantly fewer DREB genes. For example, rice has 57 [44] while the

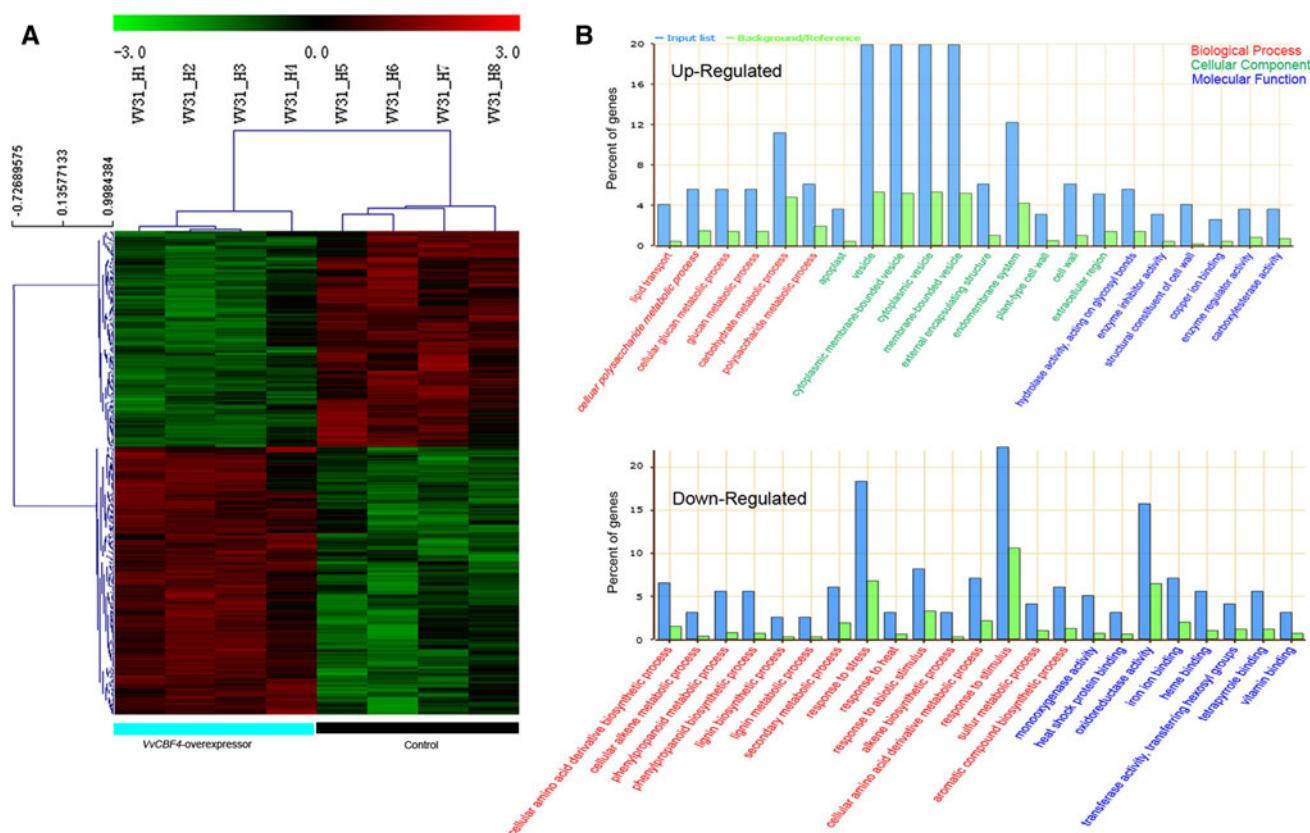


Fig. 5 **a** Hierarchical clustering of differentially expressed genes between *VvDREB19* (*VvCBF4*)-overexpressor and empty-vector control. **b** Gene ontology of significantly expressed genes (fold-

change > 1.5). y-Axis shows abundance percentage for GO term; x-axis presents definitions of GO terms. The scale bar represents the log 2 value from -3.0 to 3.0

model system *A. thaliana* has 56 [4]. Moreover, Wang et al. [45] have identified 74 NAC genes in the 12× grapevine genome whereas *A. thaliana* has 96. This difference may be due to gene evolution and duplication.

Each of the 38 *VvDREBs* identified here has an AP2 DNA-binding domain. Amino acids in that domain show commonality as well as particularity among subgroups, which may influence their capacity to bond to the DNA double helix (Figs. 1, 2). A MEME investigation of motifs outside that domain has revealed some that are group-specific, e.g. Motif 8-12 (Fig. 2). However, further research is needed to determine whether they have any biological function.

Microarray data for grapevine shared online is a valuable resource when investigating gene expression. Here, 19 *VvDREBs* (2 for *VvDREB1*, 1 for *VvDREB2*, 7 for *VvDREB4*, 5 for *VvDREB5*, and 4 for *VvDREB6*) were identified from two types of Affymetrix grapevine arrays (Fig. 3). Each displays a diverse pattern of expression under various treatments.

Both *VvDREB28* and *VvDREB38* play important roles in tolerance to drought and salt; *VvDREB30* is significantly up-regulated in response to drought, salt, cold, heat, or

oxidative stress; and *VvDREB13*, 16, and 25 are strongly induced by cold, and are also responsive to salt and oxidative stresses (Fig. 3a). *VvDREB02* is responsive only to cold and heat, and *VvDREB13* is extraordinarily sensitive to fluctuations in temperature, being dramatically up-regulated by more than 20-fold and 120-fold under cold and heat, respectively (Fig. 3b). By comparison, most *VvDREBs* show little response or downregulation after ABA application (Fig. 3a). However, *VvDREB27* can be highly induced (over 15-fold) after 3 days of such treatment and can also be regulated by both long and short photoperiods (Fig. 3a).

It is worth noting that an ERF-associated amphiphilic repression (EAR)-like motif (KPDNLNxxP) exists in two A5 members, *VvDREB08* and *VvDREB26*. These EAR motif-containing repressor proteins play a key role in modulating plant defenses and stress responses by inhibiting the expression of DREB TFs [46, 47]. That is, they can suppress defense and stress-inducible genes in the absence of any stress. This is confirmed from microarray analysis. Transcript levels for both are hardly changed under drought, salt, cold, heat, or oxidative stress but are up-regulated during the fruit ripening process (Fig. 3a).

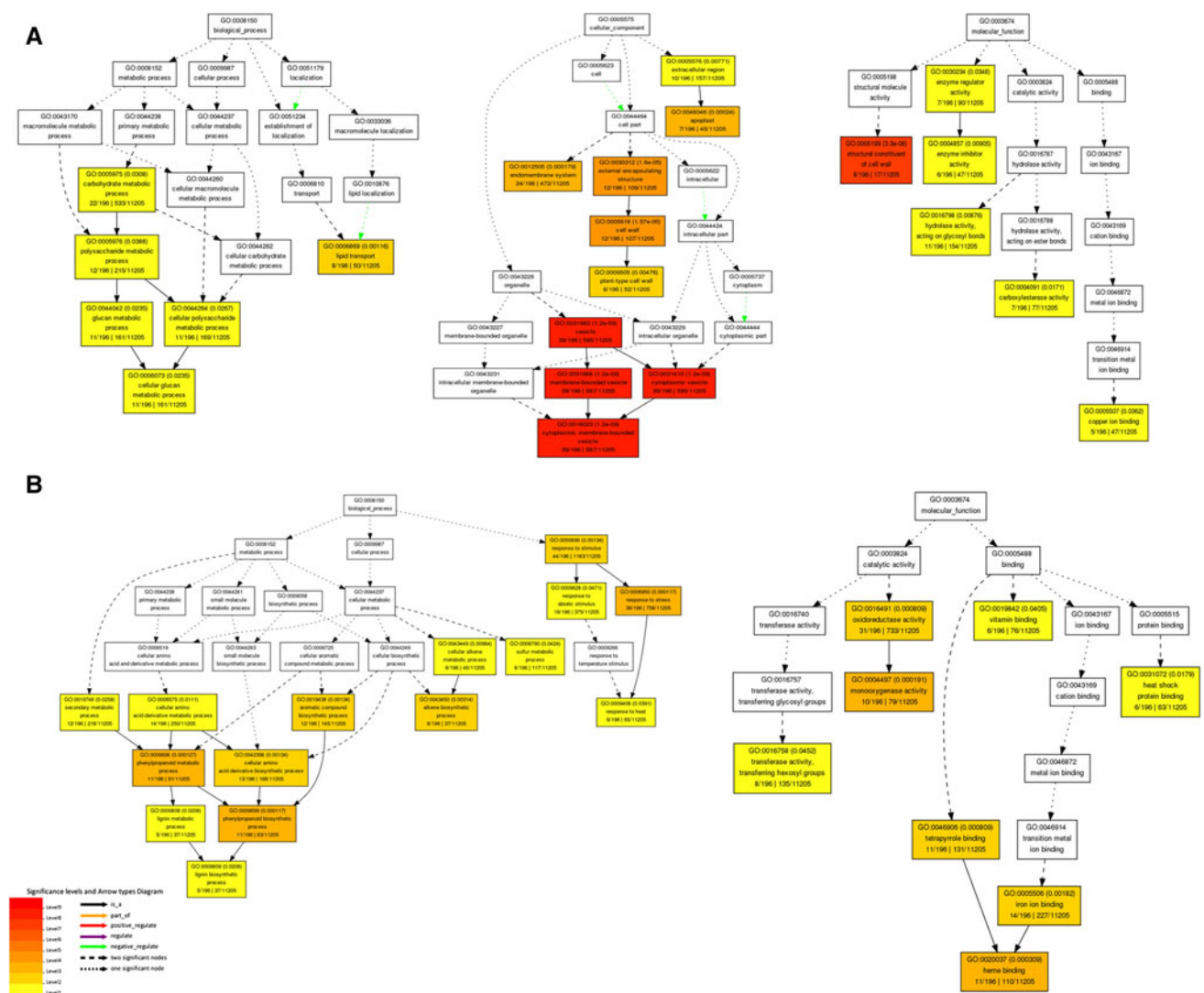


Fig. 6 Hierarchical tree of significant GO terms in categories of biological processes, cellular components, and molecular functions, generated by SEA. GO IDs, term definitions, and statistical

information are shown in each box. Different shades of colors represent degree of significance, from level 1–9. Arrow types are described in lower-left corner

VvDREB26 is also responsive to light. One explanation for this might be that they are constitutively expressed under non-stress conditions, suppressing defense and stress-responsive genes. However, under abiotic stress, their repressive effect is removed so that stress-related genes function instead.

Expression of *VvDREB12*, *16*, and *20*, all A4 members, is markedly decreased during fruit-ripening, with this downregulation being more obvious in the skin than in the flesh (Fig. 3b). This is especially true for *VvDREB20* where transcript levels do not change in flesh samples but are drastically altered in the skin (Fig. 3b). Some explanation might be gained by examining their expression profiles during the stages of bud development. In previous evaluations, transcripts of *VvDREB13*, *14*, and *16* were elevated

only in June and July, when temperatures were highest. However, from September to November, those plants were in an endodormancy period, when they had not yet entered deep dormancy. *VvDREB12* was induced only in that period and *VvDREB27* was also significantly up-regulated (>30-fold) at that time, indicating that both are involved in this pre-dormancy induction phase (Fig. 3b).

Regulatory *cis*-elements were also examined through a preliminary promoter analysis. The presence of MYB-binding sites, W-box, and ABRE suggests that *VvDREBs* are regulated by TFs such as MYB, and/or WRKY, and/or ABRE-binding proteins. Moreover, we found a DRE *cis*-element in the promoters of *VvDREB28* and *VvDREB31*, implying that both members maybe regulated by DREB TFs (Fig. 4).

Microarray data for the *VvDREB23* (*VvCBF4*)-overexpressor in unstressed plants as well as the empty-vector control transformant obtained online provided us with an opportunity to examine the global gene changes caused by *VvDREB1*s. After the original data was downloaded and re-analyzed for comparisons, we discovered that the numbers of significantly expressed genes (fold change > 1.5, p value < 0.05) were quite different from those previously reported (Fig. 5). For example, we found 248 up-regulated genes and 229 down-regulated genes versus 47 and 12, respectively. This discrepancy may be due to the application of different statistical test methods, i.e. traditional t tests were used previously whereas our calculations were made via LIMMA, which is an advanced program for statistical analysis. As shown in the GO enrichment, *VvDREB23* directly or indirectly up-regulates genes that encode proteins with roles in sugar metabolism, lipid transport, development of the cytoplasmic membrane, and cell wall biogenesis. All of these processes can determine the level of cold resistance manifested by a plant. However, we also found that constitutive overexpression of *VvDREB23* can greatly repress the expression of many genes to nearly the same extent as those that are up-regulated (i.e. changes of 1.5-fold). In particular, those genes encode heat shock proteins and are also involved in a wide range of other functions, including lignin and alkene biosyntheses, and sulfur metabolism. To the best of our knowledge, few previous reports have focused on this dichotomy of functions. For example, some genes might be down-regulated for heat shock proteins and lignin biosynthesis when plants are under stress. However, for the *VvDREB23*-overexpressor, plants that are not subjected to stress might have an intrinsic, enhanced capacity for stress resistance, so that additional effort is not required for sustained activation of such genes.

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